ALEKSANDRA KOWALSKA

Graduate Software Engineer | Full-Stack Developer | Creative Technologist
075 345 44 116 | akreative@outlook.com | Southampton, UK |
https://github.com/akreative-dev | https://www.akreative-dev.co.uk |

PROFESSIONAL PROFILE:

Computer Science MSc graduate with Distinction, combining software development, data science, machine learning and graphic design. I enjoy building things, understanding how they work and figuring out how to make them work better.

I have independently developed full-stack applications, database systems and data-processing pipelines using Python, Flask, PostgreSQL, PHP and JavaScript. My main MSc project, FreeFrom14, involved building an NLP-powered recipe search engine using semantic retrieval, spaCy NER, Word2Vec and relational database design.

Alongside my technical work, I bring years of experience working in a busy healthcare environment where accuracy, organisation, communication and staying calm under pressure are part of the job. I am a practical problem-solver who enjoys understanding challenges from different perspectives and finding thoughtful, workable solutions.

I am looking for a graduate or junior software engineering role where I can keep learning, contribute to real projects and build useful, well-considered software.

TECHNICAL SKILLS:

- **Languages:** Python · JavaScript · PHP · R · SQL · HTML5 · CSS3
- **Web & Frameworks:** React · Next.js · Flask · Tailwind CSS · Responsive Design
- **Data & Machine Learning:** PostgreSQL · Scikit-learn · TensorFlow/Keras · NLP · CNNs · NER
- **Development:** Git · GitHub · VS Code · Testing · Validation · Security Fundamentals
- **Design:** UI/UX · Information Architecture · Typography · Adobe Creative Suite

SELECTED PROJECTS:

FreeFrom14 – NLP-Powered Allergen-Aware Recipe Search Engine | *MSc Computer Science*

MSc Computer Science · Python · Flask · PostgreSQL/Supabase · spaCy · Word2Vec

- Independently designed and developed an end-to-end recipe search system, covering data acquisition, processing, NLP, database architecture, backend development and deployment.
- Built a **hybrid NLP pipeline** combining Regex-based rules, spaCy Named Entity Recognition and Word2Vec to identify ingredients and support allergen filtering and substitution.
- Designed a normalised PostgreSQL data model and deployed the application using Flask, PythonAnywhere and Supabase, while evaluating ethical data sourcing and API constraints.

WearView Academy – IT Support Management System

PHP · SQL · JavaScript · HTML5 · CSS3

- Designed and developed a multi-page IT support management application with a relational SQL database supporting issue creation, retrieval, updating and completion workflows.
- Implemented authentication and access controls, then tested functionality, accessibility, responsiveness and security, including SQL injection vulnerabilities.

ADDITIONAL · ML · PROJECTS

CIFAR-10 Image Classification — CNN & Transfer Learning

Python · TensorFlow/Keras · CNN · MobileNetV2

- ★ Investigated CNN architecture, dropout, L2 regularisation, data augmentation and transfer learning. Achieved 76% accuracy using MobileNetV2, with experiments demonstrating the benefits of pre-trained representations and regularisation under constrained computational resources.

Breast Cancer Classification — Supervised Machine Learning

Python · Scikit-learn

- ★ Compared KNN, Naive Bayes, Decision Tree and Neural Network classifiers. KNN achieved **98.25% accuracy** on the selected dataset.

Bank Marketing — Artificial Neural Network

Python · Neural Networks

- ★ Investigated class imbalance, oversampling and hidden-layer size when predicting term-deposit subscriptions. A **12-node hidden layer** produced the strongest overall performance.

EDUCATION:

- ★ **MSc Computer Science** | *Grade: Distinction* | University of Sunderland | Graduated: June 2026
- ★ **BA (Hons) Graphic Arts: Graphic Design** | *Grade: 2:1* | University of Southampton | Graduated: July 2014
- ★ **BTEC National Diploma in Art and Design: Graphic Design** | Grade: Distinction | Northampton College | Graduated: July 2011

OPERATIONAL · EXPERIENCE:

Ward Clerk — Acute Emergency General Surgery | *Southampton General Hospital* | Aug 2018 - Present

- ★ Work in a fast-paced emergency surgical environment, managing patient records, admissions, transfers and live clinical queues where accuracy and timely communication are essential.
- ★ Handle sensitive information within strict data protection and healthcare procedures while coordinating communication between clinical teams, patients and families.
- ★ Collected and tracked patient admission and outcome data that contributed to planning the expansion of the unit and the introduction of a new Same Day Emergency Care room.

Housekeeper & Hostess | *Southampton General Hospital* | Sept 2014 - Aug 2018

- ★ Worked within a busy healthcare environment, maintaining high standards of cleanliness, safety, organisation and day-to-day ward logistics.

Industrial Printer / Screen Technician | *Ritter UK (Innovations in Plastics)* | Sept 2007 - Aug 2009

- ★ Operated and calibrated industrial silk-screen machinery and carried out fault-finding and adjustments during production.
- ★ Worked with precise colour mixing and supported the training of new team members.

LANGUAGES & OTHER · TOOLS:

* Languages: English (Fluent) | Polish (Native)
* Tools: VS Code | Git | GitHub | Adobe Creative Suite | Microsoft Office